AI for Scientific Writing

Session 4 · Reference management with Mendeley — and a first step into LaTeX

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Tecnológico de Monterrey · Semana Tec · online (Google Meet)

Yesterday in one minute — and today

Yesterday

- fluent $\neq$ true; declaring is not enough, you must verify
- AI is neither an author nor a source
- group principles and the responsible-use letter

Today 18:00: letter (Act. 10), Tasks 5 and 6.

Today

07:10	Reference management: why and how
07:40	Activity 11 — your Mendeley library
08:20	Break
08:30	Activity 12 — citing in Word with Mendeley Cite
09:20	LaTeX and BibTeX in 20 minutes
09:45	Tasks 7 and 8

Why a reference manager

By hand:

- misspelt authors, wrong years
- every journal wants a different style
- citations in the text missing from the list (and vice versa)
- switching from APA to IEEE = redoing everything

With a manager:

- one database with your sources and their PDFs
- citations and bibliography generated automatically
- change style with one click
- your notes and highlights in one place

But: the manager copies the data you give it. If a field is wrong, the citation is wrong.

The Mendeley toolkit



- **Mendeley Desktop** can no longer be downloaded since 1 September 2022: use **Mendeley Reference Manager**.
- Free, from Elsevier. Free open-source alternative: **Zotero** (also works in Google Docs).

Install and set up (10 min)

1. Create your account at `mendeley.com` (use your Tec email).
2. Download and install **Mendeley Reference Manager** (Windows, macOS or Linux) — or use the web version.
3. Install the **Mendeley Web Importer** in Chrome, Edge or Firefox.
4. In Word: *Insert* → *Add-ins* → search **Mendeley Cite** and sign in.

Mendeley Cite works in Word 2016 or later and in Word online (Office 365), even when Reference Manager is closed.

Adding references — four ways

1. **Web Importer:** on the article page (Scopus, IEEE Xplore, ScienceDirect, Google Scholar), click the extension button.
2. **Drag the PDF** into Reference Manager: it extracts title, authors and DOI.
3. **Import** a `.bib` or `.ris` file exported from Scopus or Google Scholar.
4. **By hand**, when there is no other option.

Always check the metadata:

- ☐ complete, correctly spelt authors
- ☐ exact title
- ☐ journal or conference, volume, pages
- ☐ year
- ☐ a DOI that opens
- ☐ correct type (article, conference, book)

Organise and read

- **Collections:** one per project ("Artículo Semana Tec").
- **Tags:** topic, method, "verified".
- **Highlights and notes** on the PDF.
- **Notebook:** gathers highlights from several PDFs — a draft of your literature matrix.
- **Private groups** to share sources.

Good practice

- one manager per document
- don't edit the bibliography by hand: it is lost on update
- keep the PDF you read with the reference
- if AI suggests a source, it enters the manager only after you open it

Activity 11 — your Mendeley library (40 min)

1. Create the collection **Artículo Semana Tec**.
2. Add your **5+ sources from Activity 5** (Web Importer, PDF or `.bib`).
3. Check and fix the metadata with the list.
4. Write one note per source: the main idea in one line.
5. Tag as `verified` the ones you opened and read.

When done: share your screen for 1 minute or post a screenshot in the chat.

Question: which field was wrong? We discuss it together.

Citing in Word with Mendeley Cite

1. Put the cursor where the citation goes.
2. Mendeley Cite → search the source → **Insert citation**.
3. At the end of the document: **Insert bibliography**.
4. Change the style: **APA 7th ed.** or **IEEE** — everything updates.
5. If you fix a field in Mendeley, **refresh** the citations.

One source, two styles

APA 7: . . . invent references (Walters & Wilder, 2023).

Walters, W. H., & Wilder, E. I. (2023). Fabrication and errors in the bibliographic citations generated by ChatGPT. *Scientific Reports, 13*, 14045. https://doi.org/10.1038/s41598-023-41032-5

IEEE: . . . invent references [1].

[1] W. H. Walters and E. I. Wilder, "Fabrication and errors in the bibliographic citations generated by ChatGPT," *Sci. Rep.*, vol. 13, Art. no. 14045, 2023, doi: 10.1038/s41598-023-41032-5.

Activity 12 — the manager in action (50 min)

In your article (Task 4/6 draft):

1. Insert **at least 5 citations** with Mendeley Cite.
2. Insert the **bibliography**.
3. Switch to **IEEE** and then to **APA 7** — check that everything changes.
4. Check: every citation in the text is in the list and every source in the list is cited.
5. Fix any error in Mendeley and refresh.

Typical errors

- citation pasted as text (does not update)
- "et al." where the style wants all authors
- two versions of the same source (duplicate)
- title in all capitals

Why LaTeX?

- the standard in engineering, physics and mathematics
- IEEE, Springer and Elsevier provide LaTeX templates
- equations, figures and tables number themselves
- automatic bibliography with **BibTeX**
- the template sets the format; you write content

Overleaf (`overleaf.com`): LaTeX in the browser, free, nothing to install, online collaboration.

Flow:
Mendeley → export `refs.bib` → Overleaf → `\cite{key}` → PDF with references.

A minimal LaTeX document

```
\documentclass[11pt]{article}
\usepackage[english]{babel}
\title{Your article title}
\author{Your name}
\begin{document}
\maketitle
\begin{abstract} Abstract... \end{abstract}
\section{Introduction}
LLMs invent references~\cite{walters2023fabrication
    }.
\begin{equation}
  \eta = P_{out} / P_{in}
\end{equation}
\bibliographystyle{ieeetr}
\bibliography{refs}   % refs.bib
\end{document}
```

- `\section` numbers itself
- `equation` numbers the equation
- `\cite{key}` gives [1]
- `\bibliography{refs}` builds the list from what you cite
- swap `ieeetr` for another style and everything reformats

BibTeX: from Mendeley to Overleaf

1. In Mendeley, select your collection.
2. **File** → **Export** → **BibTeX (.bib)**.
3. In Overleaf, upload the file as `refs.bib`.
4. Copy each entry's **key** and use it in `\cite{}`.
5. Compile: Overleaf runs BibTeX for you.

```
@article{walters2023fabrication,
  author  = {Walters, William H. and
             Wilder, Esther Isabelle},
  title   = {Fabrication and errors in
    the bibliographic citations
    generated by {ChatGPT}},
  journal = {Scientific Reports},
  volume  = {13},
  pages   = {14045},
  year    = {2023},
  doi     = {10.1038/s41598-023-41032-5}
}
```

Demo — your article in Overleaf (10 min)

1. Download the **course template** (course site, session 4): `main.tex` + `refs.bib`.
2. Overleaf → **New project** → **Upload project** → the `.zip`.
3. **Recompile**: references [1] and [2] appear.
4. Replace `refs.bib` with the one you exported from Mendeley and cite one of your sources.
5. Optional: open the official IEEE template in the Overleaf gallery.

Submitting in LaTeX is **optional** this week; Word with Mendeley Cite is enough.

Tasks 7 and 8 — independent work

Task 7 — reference manager practice (5 %)
Your references with DOI in Mendeley (or Zotero), organised and cited in the required style.

Task 8 — reference review (5 %)
Check every reference's fields and enter by hand those the manager got wrong.

Dates

- Letter (Act. 10), Tasks 5 and 6: **today, 18:00**
- Tasks 7 and 8: **Friday 30, 18:00**

Tomorrow: final presentations (Act. 13, 15 %). Bring your article with its references.

A source stored well

is a citation that never breaks.



lrodriguez15.github.io/ai-scientific-writing.html